Algebra 1B

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1 Vector Spaces and Linear Maps

Definition. A **vector space** V over a field $\mathbb{F}$ is a set V with two operations, $+$ and $\times$.

All algebraic manipulation in a vector space ultimately comes from the Vector Space Axioms and Field Axioms.

1.1 Subspaces

Definition. Let V be a vector space over $\mathbb{F}$. A subset $U \subseteq V$ is a (linear) **subspace** if

- $0 \in U$
- U is closed under linear combinations, meaning

$$\forall u, w \in U,\ \lambda, \mu \in \mathbb{F} \quad \lambda u + \mu w \in U.$$

In this case, we write $U \leq V$.

Remark. Because addition and multiplication is defined, U is a vector space in its own right.

Any V has two special subspaces, V itself and the trivial subspace $\{0\}$. A subspace U with $U \neq V$ is said to be 'proper'.

1.2 Linear Maps

Definition. Let V, W be vector spaces over $\mathbb{F}$. A map $\phi : V \to W$ is a **linear map** if

$$\phi(\lambda v + \mu w) = \lambda\phi(v) + \mu\phi(w) \quad \lambda, \mu \in \mathbb{F},\ v, w \in V.$$

If $V = W$, then ϕ is a linear operator on V.

Lemma.

- The identity map $\mathrm{id}_V : V \to V$, defined $\mathrm{id}_V(v) = v$, is a linear operator over V.
- If $\phi : V \to W$ is a linear map then

$$\phi(0) = 0, \quad \phi(-v) = -\phi(v).$$

- If $\phi : V \to W$, $\psi : W \to U$ are linear maps then $\psi \circ \phi : V \to U$ is also a linear map
- If $\phi : V \to W$ is a linear bijecion then $\phi^{-1} : W \to V$ is also linear.

Proof.

- $\mathrm{id}_V(\lambda v + \mu w) = \lambda v + \mu w = \lambda \mathrm{id}_V(v) + \mu \mathrm{id}_V(w)$, hence id_V is a linear operator.
- $\phi(0) = \phi(v + (-v)) = \phi(v) - \phi(v) = 0$
- $\psi \circ \phi(\lambda v + \mu w) = \psi(\lambda \phi(v) + \mu \phi(w)) = \lambda \psi \circ \phi(v) + \mu \psi \circ \phi(w)$ hence $\psi \circ \phi$ is linear
- Let $v, w \in W$. Hence $\phi(\phi^{-1}(w_i)) = w_i$ for arbitrary w_i.

$$\begin{aligned} \lambda v + \mu w &= \lambda \phi(\phi^{-1}(v)) + \mu \phi(\phi^{-1}(w)) \\ &= \phi\big(\lambda \phi^{-1}(v) + \mu \phi^{-1}(w)\big). \end{aligned}$$

  Then, by applying ϕ^{-1} to both sides

$$\begin{aligned} \phi^{-1}(\lambda v + \mu w) &= (\phi^{-1} \circ \phi)(\lambda \phi^{-1}(v) + \mu \phi^{-1}(w)) \\ &= \lambda \phi^{-1}(v) + \mu \phi^{-1}(w). \end{aligned}$$

□

Definition. A linear bijection $\phi : V \to W$ is called a (linear) **isomorphism**.

We say V is **isomorphic to** W if an isomorphism $\phi : V \to W$ exists, written $V \cong W$.

1.3 Subspaces from Linear Maps

Definition. Let $\phi : V \to W$ be linear.

The **kernel** of ϕ is $\ker \phi \subseteq V$ given

$$\ker \phi = \{v \in V \mid \phi(v) = 0\}.$$

The **image** of ϕ is $\operatorname{Im} \phi \subseteq W$ given by

$$\operatorname{Im} \phi = \{\phi(v) \mid v \in V\}$$

Proposition. Let $\phi : V \to W$ be linear. Then

- $\ker \phi \leq V$
- $\operatorname{Im} \phi \leq W$.

Proof.

- Let $v, w \in \ker \phi$. Then

$$\phi(\lambda v + \mu w) = \lambda \phi(v) + \mu \phi(w) = 0$$

  hence $\lambda v + \mu w \in \ker \phi$, so $\ker \phi \leq V$.

- Let $\phi(v), \phi(w) \in \operatorname{Im} \phi$. Then

$$\lambda \phi(v) + \mu \phi(w) = \phi(\lambda v + \mu w) \in \operatorname{Im} V$$

  Hence, $\operatorname{Im} V \leq V$.

□

Proposition. A linear map $\phi : V \to W$ is injective if and only if $\ker \phi = \{0\}$

Proof.
($\Rightarrow$) For any linear map, $\phi(0) = 0$ and 0 is always in the kernel. Hence if $\phi(v) = 0 = \phi(0)$, by injectivity $v = 0$, so $\ker \phi = \{0\}$.
($\Leftarrow$) Suppose $\phi(v) = \phi(w)$, therefore $\phi(v - w) = 0$ by linearity and since $\ker \phi = \{0\}$, $v - w = 0$, meaning ϕ is injective. □

Corollary. A linear map $\phi : V \to W$ is an isomorphism if and only if $\ker \phi = \{0\}$ and $\operatorname{Im} \phi = W$.

2 Bases, Coordinates and Matrices

2.1 On Lists

Definition. A **list** of vectors in a vector space V is a finite sequence $v_1, ..., v_n$ of elements of V. A **sequence** is a map $\{1, ..., n\} \to V$ defined $i \to v_i$

There are some differences between lists and (sub)sets:

- lists are ordered
- lists allow the same element to appear more than once

2.2 Spanning and Independent lists

Definition. Let $\alpha : v_1, ..., v_n \in V$ be a list of vectors in a vector space V over $\mathbb{F}$.

The **span** of α is given by

$$span\ \alpha = span\{v_1, ..., v_n\} := \left\{ \sum_{i=1}^{n} \lambda_i v_i | \lambda_1, ..., \lambda_n \in \mathbb{F} \right\} \leq V.$$

More generally, for $S \subseteq V$, the span of S is

$$spanS := \left\{ \sum_{i=1}^{n} \lambda_i s_i | s_1, ..., s_n \in S, \lambda_1, ..., \lambda_n \in \mathbb{F}, n \in \mathbb{N} \right\} \leq V.$$

$span\ S$ is the smallest subspace of V containing S :

$$span\ S = \bigcap_{u \leq V\ |\ S \subseteq u} u.$$

Definition. Let $\alpha : v_1, ..., v_n \in V$ be a list of vectors in a vector space V over $\mathbb{F}$.

- α is **(linearly) independent** if, whenever $\lambda_1, ..., \lambda_n \in \mathbb{F}$ have

$$\sum_{i=1}^{n} \lambda_i v_i = 0 \text{ then } \lambda_1 = ... = \lambda_n = 0.$$

  and **(linearly) dependent** otherwise.

- α is **spanning** or **spans** if

$$span\ \alpha = V.$$

- α is a **basis** of V if it is both independent and spans.

Remark. Independence of a list $\alpha : v_1, ..., v_n \in V$ says that no element $v_i \in \alpha$ can be expressed as another element multiplied by a scalar λ, or furthermore

$$\sum_{i=1}^{n} \lambda_i v_i \neq \sum_{i=1}^{n} \mu_i v_i.$$

unless $\lambda_i = \mu_i$ for all i, meaning there is a one-to-one mapping from the list $\lambda_1, ..., \lambda_n$ to the span of α.

Lemma.

- A single vector v is independent if $v \neq 0$
- A two element list v_1, v_2 is dependent $\Leftrightarrow v_1 = \lambda v_2$ for some $\lambda \in \mathbb{F}$

2.3 Spanning and Independence in $\mathbb{F}^n$

Proposition. Let $\alpha : v_1, ..., v_n \in \mathbb{F}^M$ and define $A \in M_{M\times N}(\mathbb{F})$ by $col_i(A) = v_i$. Let $\hat{A}$ be an REF matrix obtained from A by gaussian elimination. Then

- The list α is independent if and only if every column of $\hat{A}$ has a pivot.
- α is spanning if $\hat{A}$ has no zero rows.

Proof.

- α is independent $\Leftrightarrow \sum_{i=1}^n \lambda_i v_i \neq 0$ for nonzero vector $\lambda \Leftrightarrow N_A = \{0\}$ $\Leftrightarrow \hat{A}$ has a pivot in every column
- α spanning $\Leftrightarrow \forall v, v = \sum_{i=1}^n \lambda_i v_i$ for some vector $\lambda \Leftrightarrow Ax = b$ always has a solution $\Leftrightarrow A$ has no zero rows

□

2.4 Bases and Isomorphisms

Proposition. Let $\alpha : v_1, ..., v_n$ be a list in V and define $\phi_\alpha : \mathbb{F}^n \to V$

$$\phi_\alpha(\mathbf{x}) \to \sum_{i=1}^n v_i x_i.$$

Then

- α is independent if and only if ϕ_α is injective
- α spans if and only if ϕ_α is surjective.

Hence α is a basis if and only if ϕ_α is an isomorphism.

Proof.

- α independent $\Leftrightarrow \phi_\alpha(v) = 0 \Rightarrow v = 0 \Leftrightarrow \phi_\alpha(v - w) = 0 \Rightarrow v - w = 0$ $\Leftrightarrow \phi_\alpha(v) = \phi_\alpha(w) \Rightarrow v = w \Leftrightarrow \phi_\alpha$ injective.
- α spans $\Leftrightarrow$ every v can be written in terms of $\alpha \Leftrightarrow \forall w \exists v \phi_\alpha(v) = w$ $\Leftrightarrow \phi_\alpha$ surjective.

□

Corollary. A list $\alpha : v_1, ..., v_n \in V$ is a basis if and only if every $v \in V$ can be written as a linear combination

$$v = \sum_{i=1}^n x_i v_i$$

for unique $x_1, ..., x_n \in \mathbb{F}$.

Proposition. Let $\phi : \mathbb{F}^n \to V$ be an isomorphism. Then $\phi = \phi_\alpha$ where $\alpha : \phi(e_1), ..., \phi(e_n) \in V$ is a basis.

Proof.

$$\phi(v) = \phi\left(\sum_{i=1}^{n} \lambda_i e_i\right) = \sum_{i=1}^{n} \lambda_i \phi(e_i) = \phi_\alpha(v)$$

□

2.5 Bases and Coordinates

A basis allows us to represent abstract vectors by lists of scalars (coordinates).

Definition. Let $\alpha : v_1, ..., v_n \in V$ be a basis of V. For $v \in V$, we call the vector $(x_1, ..., x_n) \in \mathbb{F}^n$ defined by

$$v = \sum_{i=1}^{n} x_i v_i.$$

This is the **coordinate vector** of v with respect to α.

Remark.

- $v = \phi_\alpha(x_1, ..., x_n)$ so

$$(x_1, ..., x_n) = \phi_\alpha^{-1}(v).$$

- Reordering α gives a different basis with different coordinate vectors.

2.6 Bases, Linear Maps and Matrices

Definition. Let V, W be vector spaces over $\mathbb{F}$, $\alpha : v_1, ..., v_n \in V$ be a basis of V, $\beta : w_1, ..., v_m \in W$ be a basis of W, and $\phi : V \to W$ is a linear map.

The **matrix** of ϕ with respect to α, β is $A \in M_{m,n}(\mathbb{F})$ given by

$$\phi(v_j) = \sum_{i=1}^{m} A_{ij} w_i.$$

This means the jth column of A is $\phi(v_j)$ written in terms of β.

When $V = W$, we can take $\alpha = \beta$

Definition. Let V be a vector space over $\mathbb{F}$ with basis $\alpha = v_1, ..., v_n$ and $\phi : V \to V$ is a linear map on V. The **matrix** of ϕ with respect to α is

$A \in M_n(\mathbb{F})$ given by

$$\phi(v_j) = \sum_{i=1}^{n} A_{ij} v_i.$$

Proposition. Let $\alpha : v_1, ..., v_n$ be a basis of V, $\beta : w_1, ..., v_m$ be a basis of W, and $\phi : V \to W$ is a linear map with matrix A with respect to α, β.

- For $v \in V$ with coordinate vector x with respect to α, $\phi(v)$ has coordinates y with respect to β where

$$\underline{y} = A\underline{x}.$$

- We have

$$\phi = \phi_\beta \circ \phi_A \circ \phi_\alpha^{-1}$$

  or, equivalently

$$\phi_A = \phi_\beta^{-1} \circ \phi \circ \phi_\alpha.$$

Proof.

- To prove the first part:

$$\begin{aligned}
\phi(v) &= \sum_{j=1}^{n} x_j \phi(v_j) \\
&= \sum_{j=1}^{n} x_i (\sum_{i=1}^{m} A_{ij} w_i) \\
&= \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} x_j w_i \\
&= \sum_{i=1}^{m} (A\underline{x})_i w_i.
\end{aligned}$$

  hence $\phi(v)$ has coordinates $A\underline{x}$ with respect to β, so $y = A\underline{x}$.

- Let $w = \phi(v)$. Since $y = \phi_\beta^{-1}(w) = A\phi_\alpha^{-1}(v)$ we have

$$\phi_\beta^{-1} \circ \phi = \phi_A \circ \phi_\alpha^{-1}$$

  from which we can obtain the necessary equations.

□

Proposition. Let V, W, U be vector spaces over $\mathbb{F}$ with bases α, β, γ respectively. Let $\phi : V \to W$ and $\psi : W \to U$ be linear maps with matrices A with respect to α, β and B with respect to β, γ.

Then $\psi \circ \phi$ has matrix BA with respect to α, γ.

Proof.

$$\begin{aligned}\psi \circ \phi &= (\phi_\gamma \circ \phi_B \circ \phi_\beta^{-1}) \circ (\phi_\beta \circ \phi_A \circ \phi_\alpha^{-1}) \\ &= \phi_\gamma \circ \phi_B \circ (\phi_\beta^{-1} \circ \phi_\beta) \circ \phi_A \circ \phi_\alpha^{-1} \\ &= \phi_\gamma \circ \phi_B \circ \phi_A \circ \phi_\alpha^{-1} \\ &= \phi_\gamma \circ \phi_{BA} \circ \phi_\alpha^{-1}.\end{aligned}$$

hence $\psi \circ \phi$ has matrix BA with respect to α, γ.

□

2.7 Change of Basis

What happens if we want to compose two linear maps that use different bases?

Definition. Let $\alpha : v_1, ..., v_n$ and $\alpha' : v_1', ..., v_n'$ be two bases of V. Then the change of basis matrix with respect to α, α' is $P \in M_n(\mathbb{F})$ such that

$$v_j' = \sum_{i=1}^{n} P_{ij} v_i.$$

This means the jth column of P is v_j' written in terms of α.

Lemma. $\phi_P = \phi_\alpha^{-1} \circ \phi_{\alpha'}$ and so is an isomorphism, hence P is invertible.

Proof. ϕ_P is the identity map with respect to α', α, hence

$$\begin{aligned}\phi_P &= \phi_\alpha^{-1} \circ \mathrm{id}_V \circ \phi_{\alpha'} \\ &= \phi_\alpha^{-1} \circ \phi_{\alpha'}.\end{aligned}$$

and as it is the composition of isomorphism, it is itself and isomorphism and hence invertible. □

Corollary. The change of basis matrix from α' to α is P^{-1}.

Proposition. Let P be a change of basis matrix from α to α'. Let $v \in V$ have coordinate vector x with respect to α and x' with respect to α'. Then

$$\underline{x} = P\underline{x}'.$$

Proof. P is the matrix of id_V with respect to α' and α, so $\mathrm{id}_V = \phi_\alpha \circ \phi_P \circ \phi_{\alpha'}^{-1}$ or $\phi_P = \phi_\alpha^{-1} \circ \phi_{\alpha'}$. Since

$$\phi_\alpha(\underline{x}) = \phi_{\alpha'}(\underline{x}')$$

this means

$$\begin{aligned}\underline{x} &= \phi_\alpha^{-1} \circ \phi_{\alpha'}(\underline{x}') \\ &= P\underline{x}'.\end{aligned}$$

□

Proposition. Let α, α' be bases of V with change of basis matrix P from α to α'. Let β, β' be bases of W with change of basis matrix Q from β to β'.

Let $\phi : V \to W$ is a linear map with matrix A with respect to α, β and A' with respect to α', β'. Then

$$A' = Q^{-1}AP.$$

Proof. We can say $\phi = \mathrm{id}_W \circ \phi \circ \mathrm{id}_V$, with matrices Q^{-1}, A and P respectively. Therefore $A' = Q^{-1}AP$. □

Corollary. If α, α' are bases of V with change of basis matrix P from α to α', then, if $\phi : V \to W$ is a linear operator on V with matrix A with respect to α and A' with respect to α', then

$$A' = P^{-1}AP.$$

These results prompt the following definition.

Definition.

- Matrices $A, A' \in M_{M,N}(\mathbb{F})$ are **equivalent** if there exists invertible matrices $Q \in M_M(\mathbb{F}), P \in M_N(\mathbb{F})$ such that

$$A' = Q^{-1}AP.$$

- $A, A' \in M_N(\mathbb{F})$ are **similar** if there exists invertible $P \in M_N(\mathbb{F})$ such that

$$A' = P^{-1}AP.$$

Both equivalence and similarity are equivalence relations.

2.8 Computations in $\mathbb{F}^n$

Example. When α is the standard mases of $\mathbb{F}^n$, and $\alpha' : v_1', ..., v_n'$ another basis, then P is easy to compute:

$$col_j(P) = v_j'.$$

The reason we care about all this: there may be a cunning choice of basis which gives a simpler matrix for a given linear map.

3 Dimension and Rank

3.1 Dimension

Lemma. (Fundamental Lemma)
Let $A \in M_{M,N}(\mathbb{F})$. If $n > m$ then there is a nonzero $x \in \mathbb{F}^n$ such that $A\underline{x} = 0$.

Theorem.

- If $v_1, ..., v_n \in V$ are an independent list of vector, and $w_1, ..., w_m$ are a spanning list, then $n \leq m$.
- If $v_1, ..., v_n$ and $w_1, ..., w_m$ are bases, then $n = m$.

Proof.

- We can write each $v_j = \sum_{i=1}^{m} A_{ij} w_i$ for some matrix $A \in M_{m,n}(\mathbb{F})$. Hence suppose
$$\sum_{j=1}^{n} \sum_{i=1}^{m} A_{ij} w_i = 0 = A\mathbf{w}$$
. If $n > 0$, there will always be a nonzero solution, contradicting the independence of $v_1, ..., v_n$. Therefore $n \leq m$.
- Since both lists are a basis so both surjective and injective, $n \leq m$ and $m \leq n$. Therefore $n = m$.

$\square$

Definition. Let V be a vector space.

- V is **finite dimensional** if it has a finite length basis and **infinite dimensional** otherwise
- If V is finite dimensional, the **dimension** of V, denoted $\dim V$ is the length of any basis of V.

Example.

- $\dim \mathbb{F}^n = n$
- $\dim \mathbb{F}^0 = \dim\{0\} = 0$
- $\dim \mathbb{F}[x] = \infty$

Dimension is preserved by isomorphism.

Proposition. Let V be a vector space over $\mathbb{F}$. Then $\dim V = n$ if and only if $\mathbb{F}^n \cong V$.

In this case, if $W \cong V$ then W is finite dimensional also with $\dim W = n$.

Proof. Any isomorphism $\phi : \mathbb{F}^n \to V$ creates a basis α of V by applying ϕ to $e_i, ..., e_n$, and is uniquely determined by this basis. The second statement can be proved by composing the two isomorphisms, meaning $W \cong \mathbb{F}^n$ and so $\dim W = n$. $\square$

Proposition.

- If $v_1, ..., v_n$ span V and $v_n \in span\{v_1, ..., v_{n-1}\}$ then $v_1, ..., v_{n-1}$ span V.
- If $v_1, ..., v_{n-1}$ is an independent list and $v_n \notin span\{v_1, ..., v_{n-1}\}$ then $v_1, ..., v_n$ is also independent.

Proof.

$$\begin{aligned}
V &= span\{v_1, ..., v_n\} \\
&= \{\sum_{i=1}^{n} \lambda_i v_i | \lambda_1, ..., \lambda_n \in \mathbb{F}, v_1, ..., v_n \in V\} \\
&= \{\sum_{i=1}^{n-1} \lambda_i v_i + v_n | \lambda_1, ..., \lambda_n \in \mathbb{F}, v_1, ..., v_n \in V\} \\
&= \{\sum_{i=1}^{n-1} \lambda_i v_i + \sum_{i=1}^{n-1} \mu_i v_i | \lambda_1, ..., \lambda_n \in \mathbb{F}, \mu_1, ..., \mu_n \in \mathbb{F}, v_1, ..., v_n \in V\} \\
&= \{\sum_{i=1}^{n-1} (\lambda_i + \mu_i) v_i | \lambda_1, ..., \lambda_n \in \mathbb{F}, \mu_1, ..., \mu_n \in \mathbb{F}, v_1, ..., v_n \in V\} \\
&= span\{v_1, ..., v_{n-1}\}.
\end{aligned}$$

For the second statement, suppose $\sum_{i=1}^{n-1} \lambda_i v_i + v_n = 0$. Then if not independent, v_n can be written as some linear combination of $v_1, ..., v_{n-1}$, so it would be in the span of those vectors, which is a contradiction. $\square$

Definition. Let α be a list of vectors in V.

- a **sublist** of α is a list obtained by deleting some elements of α
- a list β **extends** α if β is obtained by appending vectors to α.

Theorem. Let β be a finite spanning list of vectors in a vector space V extending a (possibly empty) independent list α. Then there is a basis γ of V which

- extends α
- is a sublist of β.

Proof. Let γ be the longest independent sublist of β. If all elements of β can be written in terms of γ, it is a basis. If one cannot, then appending this vector to γ gives an independent sublist of β, contradicting γ's maximal length. $\square$

Corollary.

- Any finite spanning list contains a basis
- If V is finite dimensional, any independent list can be extended to a basis.

Corollary. Let $\dim V = n$.

- Any spanning list of length n is a basis.
- Any independent list of length n is a basis.

Proof. Both lists contain a length n sublist that is a basis. Since they are themselves length n, that sublist is just the entire list. $\square$

Theorem. Let V be finite dimensional and $W \leq V$. Then W is finite dimensional also and $\dim W \leq \dim V$.

Futheremore, $\dim W = \dim V \Leftrightarrow W = V$.

Proof. Any independent list in W has length $n \leq \dim V$. Consider an independent list α of maximum length in W. If it does not span W, a longer independent list exists so there is a contradiction. Therefore, this list is a basis of W with $\dim W \leq \dim V$. If $\dim W = \dim V$, α is a basis of V with $W = span\ \alpha = V$. $\square$

Remark. If two vector spaces have the same dimension, they are only equal if one is a subspace of the other. Equal dimension alone is not enough to show they are equal.

Proposition. Let $\alpha : v_1, ..., v_n \in \mathbb{F}^m$ and define $A \in M_{m,n}(\mathbb{F})$ by $col_j(A) = \bar{v}_j$.

Let $\hat{A}$ be a REF matrix obtained from A by gaussian elimination with r pivots in cols $j_1 < j_2 < ... < j_r$. Then $v_{j_1}, ..., v_{j_r}$ is a basis for $span\ \alpha \leq \mathbb{F}^M$.

Proof. Consider the matrix $B \in M_{m,r}(\mathbb{F})$ with columns from the list $\beta : v_{j_1}, ..., v_{j_r}$. Then since EROs act on each column separately, $\hat{B}$ has a pivot in every column, thus β is independent.

To show $span\ \alpha = span\ \beta$, this is the same as proving that for every v_j, there exists x^j such that

$$B\mathbf{x^j} = \mathbf{v_j}.$$

Suppose this is unsolvable. This is trivially solvable for v_j in β, so suppose v_j is not. Then the REF of the augmented matrix $(B|v_j)$ has a row where all entries in B-columns are zero and the v_j entry is nonzero. This means this column contains a pivot, so v_j would be in β, a contradiction. $\square$

3.2 Rank

Definition. The **rank** of a linear map $\phi : V \to W$ is

$$rank\ \phi := \dim \operatorname{Im} \phi.$$

Lemma. Given $\phi : V \to W$ and isomorphisms $f : V' \to V$ and $g : W' \to W$, set $\phi' = g^{-1} \circ \phi \circ f : V' \to W'$. Then

$$rank\ \phi = rank\ \phi'.$$

Proof. Since f is surjective, $\operatorname{Im}(\phi \circ f) = \operatorname{Im} \phi$. Then restricting g to domain $\operatorname{Im} \phi$ and codomain $\operatorname{Im}(g \circ \phi)$ gives an isomorphism, hence the two images are isomorphic and so ϕ and ϕ' have the same rank. $\square$

Definition. Let $A \in M_{M,N}(\mathbb{F})$.

- The **column space** of A is the span of the columns of A.

$$colspace\ A := \left\{ \sum_{j=1}^{n} x_j col_j(A) | x_1, ..., x_n \in \mathbb{F} \right\}.$$

  The **column rank** of A is $colrank\ A := dim\ colspace\ A$.

- The **row space** of A is the span of the rows of A

$$rowspace\ A := \left\{ \sum_{j=1}^{m} x_j row_j(A) | x_1, ..., x_n \in \mathbb{F} \right\}.$$

  The **row rank** of A is $row\ rank\ A := dim\ rowspace\ A$.

- $rank\ A$ is the number of pivots in the REF of A.

Theorem. Let $A \in M_{m,n}(\mathbb{F})$. Then $rank\ A$ is equal to:

- $rank\ \phi_A$
- $colrank\ A$
- $rowrank\ A$

- The smallest number r such that

$$A = BC$$

with $B \in M_{M,r}, C \in M_{r,N}$.

Proof. Firstly, $colspace\ A = \operatorname{Im} A$ so $colrank\ A = rank\ \phi_A$. If each column of A represents a list of vectors α, the columns with pivots in $\hat{A}$ form a basis of $span\ \alpha$, so $rank\ A = colrank\ A$. It is possible to factorize $A = BC$ with $B \in M_{m,r}$ and $C \in M_{r,n}$ with $r = colrank\ A$, by setting the columns of B to be a basis of the column space. Now suppose r is not minimal, so there exists $s < r$. Then

$$col_k(A) = \sum_{j=1}^{s} c_{jk} col_j(B)$$

thus there exists a basis of length s for $colspace\ A$, which is a contradiction. Therefore r is the smallest value to factorize A. A similar argument can be used with the row rank. □

Corollary. For $A \in M_{M,N}$, $rank\ A = rank\ A^T$.

Proof. $rank\ A = rowspace\ A = colspace A^T = rank\ A^T$. □

Remark. To find the rank of a matrix A, and the 'smallest' matrices B and C such that $A = BC$, reduce A into RREF. Then the number of pivots is $r = rank\ A$. The columns of B are the columns of A where there is a pivot in $\hat{A}$, and the columns of C are the first r rows of $\hat{A}$.

Proposition.

- Let $\phi : V \to W$ be a linear map with matrix A with respect to some bases α, β. Then

$$rank\ A = \operatorname{rank} \phi.$$

- Let $A, B \in M_{M,N}(\mathbb{F})$ be equivalent. Then

$$\operatorname{rank} A = \operatorname{rank} B.$$

Proof.

- $\phi = \phi_\beta \circ \phi_A \circ \phi_\alpha^{-1}$. Then because ϕ_β and ϕ_α are isomorphisms

$$\operatorname{rank} \phi = \operatorname{rank} \phi_A = \operatorname{rank} A.$$

- Let $A = QBP^{-1}$ for invertible P, Q. Then

$$\phi_A = \phi_Q \circ \phi_B \circ \phi_P^{-1}$$

thus

$$\operatorname{rank} A = \operatorname{rank} \phi_A = \operatorname{rank} \phi_B = \operatorname{rank} B.$$

□

Remark. The above statement has a converse: If $\operatorname{rank} A = r$ then A is equivalent to $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ Additionaly, matrices of the same size and rank are equivalent under an equivalence relation.

3.3 The Rank-Nullity Theorem

Definition. The **nullity** of a linear map ϕ is

$$null\ \phi = \dim \ker \phi.$$

Theorem. (Rank-Nullity Theorem)
If V is finite dimensional and $\phi : V \to W$ is linear, then

$$rank\ \phi + null\ \phi = dim\ V.$$

Proof. Let $\dim V = n$, $\dim \ker \phi = k$. Then we must prove $\dim \operatorname{Im} \phi = n - k$. Consider a basis of $\ker \phi v_1, ..., v_k$ extended by $v_{k+1}, ..., v_n$ to form a basis of V, and we want to prove that $\phi(v_{k+1}), ..., \phi(v_n)$ is a basis of $\operatorname{Im} \phi$.

- (Independence). Suppose

$$\begin{aligned} \sum_{k+1=1}^{n} \lambda_i \phi(v_i) &= 0 \\ \phi\left(\sum_{i=k+1}^{n} \lambda_i v_i\right) &= 0 \\ \phi(w) &= 0. \end{aligned}$$

  so $w \in \ker \phi$. However this means that it could be written in terms of $v_1, .., v_k$, but this is a contradiction as the two sublists are independent due to being a basis of V.

- (Spanning). Let $v \in V$. Then

$$\begin{aligned} \sum_{i=1}^{n} \lambda_i \phi(v_i) &= v \\ \sum_{i=1}^{k} \lambda_i \phi(v_i) + \sum_{k+1=1}^{n} \lambda_i \phi(v_i) &= v \\ \sum_{k+1=1}^{n} \lambda_i \phi(v_i) &= v \end{aligned}$$

.

so $v_{k+1}, ..., v_k$ is a basis of $\operatorname{Im} \phi$. Hence

$$\operatorname{rank} \phi + \operatorname{null} \phi = \dim V.$$

□

Remark. The dimension of W need not be finite.

Theorem. Let $\phi : V \to V$ be a linear operator on finite dimensional V. Then the following are equivalent:

- ϕ is an isomorphism
- ϕ is surjective
- ϕ is injective
- $\ker \phi = \{0\}$

Proof. ϕ is surjective $\Leftrightarrow$ $\operatorname{Im} \phi = V$ $\Leftrightarrow$ $\operatorname{rank} \phi = \dim V$ $\Leftrightarrow$ $\operatorname{null} \phi = 0$ $\Leftrightarrow$ $\ker \phi = \{0\}$ $\Leftrightarrow$ ϕ is injective. □

Remark. This result fails if V is infinite dimensional.

3.4 Applications to Sums and Products

Definition. Let V, W be vector spaces over $\mathbb{F}$. Then the **product** of V and W is $V \times W$ with addition and scalar multiplication defined componentwise

$$(v, w) + (v', w') = (v + v', w + w')$$
$$\lambda(v, w) = (\lambda v, \lambda w).$$

$V \times W$ is a vector space over $\mathbb{F}$ with this addition and scalar multiplication.

Proposition. If V, W are finite dimensional, so is $V \times W$ and

$$\dim(V \times W) = \dim V + \dim W.$$

Proof. Consider a linear projection map $\pi_V : (V \times W) \to V$, $\pi_(v, w) \mapsto v$. Then $\operatorname{rank} \pi_V = \dim V$ and $\operatorname{null} \pi_V = \dim W$. Thus

$$\dim(V \times W) = \dim V + \dim W.$$

□

Definition. Let $U, W \leq V$. Then the **sum** of U and W is

$$U + W := \{u + w | u \in U, w \in W\}.$$

Remark. $U + W$ is itself a subspace of V.

Proposition. (Dimensional Formula)
If $U, W \leq V$ are finite dimensional. Then

$$\dim(U + W) + \dim U \cap W = \dim U + \dim W.$$

Proof. Consider $\phi : (U \times W) \to U + W$, $\phi(v, w) \mapsto u + w$. This clearly surjects so $\operatorname{rank} \phi = \dim(U + W)$, hence we just need to prove $\operatorname{null} \phi = \dim U \cap W$.

$$\phi(u, w) = u + w = 0.$$
$$w = -u.$$

so $w \in u$ and $u \in W$. Thus $\ker \phi \cong U \cap W$ so they have the same dimension. □

4 Determinants and Inverses

4.1 Determinants: the Sum Formula

Definition. Let $A = (a_{ij}) \in M_N(\mathbb{F})$. The determinant of A, denoted $\det A$ or $|A|$ is

$$\det A = \sum_{\sigma \in S_n} sign(\sigma) a_{\sigma(1)1} ... a_{\sigma(n)n}.$$

This is a signed product of one entry from each column, in a different row each time, all added together.

Remark. The are $n!$ summands so this formula is not easy to use as n gets larger. Fortunately, there are other ways to compute $\det A$.

Proposition. If $A = \begin{pmatrix} A' & * \\ 0 & A'' \end{pmatrix}$ with $A' \in M_k(\mathbb{F}), A'' \in M_{n-k}(\mathbb{F})$ is block diagonal (or upper triangular) then $\det A = \det A' \det A''$.

Proof. $a_{ij} = 0$ if $i > k$ and $j \leq k$, thus for the permutation to not vanish $\sigma(j) \leq k$ for all $j \leq k$. Therefore all non-zero permutations permute all $j \leq k$ to $\{1, ..., k\}$ and all $j > k$ to $\{k+1, ..., n\}$, thus every non-zero permutation can be written as the product of $\sigma' \in S_k$ and $\sigma'' \in S_{n-k}$, so the set of non-zero permutations, S'_n is equal to $S_k \times S_{n-k}$. Therefore

$$\det A = \sum_{\sigma \in S'_n} \operatorname{sign}(\sigma' \circ \sigma'') a_{\sigma'(1)1} ... a_{\sigma''(k+1)k+1} ...$$

$$\det A = \left(\sum_{\sigma \in S_k} \operatorname{sign}(\sigma') a_{\sigma'(1)1} ... \right) \left(\sum_{\sigma'' \in S_{n-k}} \operatorname{sign}(\sigma'') a_{\sigma''(k+1)k+1} ... \right)$$

$$\det A = \det A' \det A''.$$

□

Proposition. $\det A^T = \det A$.

Proof. $a_{ij} = a^T_{ji}$. Furthermore

$$\det A^T = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a^T_{\sigma(1)1} ..$$

$$\det A^T = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{1\sigma(1)} ..$$

.

But if $j = \sigma(i), i = \sigma^{-1}(j)$, so $a_{1\sigma(1)} ... a_{n\sigma(n)} = a_{\sigma^{-1}(1)1} ... a_{\sigma^{-1}(n)n}$ because $\{\sigma(1), ..., \sigma(n)\} = \{1, ..., n\}$. Additionally, since $\sigma^{-1} \in S_n$, we have $\det A^T = \det A$ as required. □

4.2 Properties of Determinants

We view $\det A$ as a function of the columns of A:
write $A = (\underline{c}_1, ..., \underline{c}_n)$ $\quad \underline{c}_j = col_j(A)$ and view $\det : (\underline{c}_1, ..., \underline{c}_n) \mapsto \det A$

Proposition. $\det A = \det(\underline{c}_1, ..., \underline{c}_n)$ is

- **multilinear** (linear in each slot separately): if $c_j = \lambda \underline{v} + \mu \underline{w}$ where v, w are column vectors, $\lambda, \mu \in \mathbb{F}$, then $\forall j, 1 \leq j \leq n$

$$\det(\underline{c}_1, ..., \underline{c}_j, ..., \underline{c}_n) = \lambda \det(\underline{c}_1, ..., \underline{v}, ..., \underline{c}_n) + \mu \det(\underline{c}_1, ..., \underline{w}, ..., \underline{c}_n)$$

- **alternating**: if $\underline{c}_i = \underline{c}_j$ for some $i \neq j$, then

$$\det(\underline{c}_1, ..., \underline{c}_n) = 0.$$

Proof.

- We have

$$a_{\sigma(1)1}...a_{\sigma(j)j}... = a_{\sigma(1)1}...(\lambda v_{\sigma(j)} + \mu w_{\sigma(j)})$$

  thus the necessary result can be obtained.

- We can split S_n into the set of even (A_n) and odd permutations. Since any odd permutations are of the form $\sigma \circ (ij)$ for some even σ, we have $S_n = A_n \cup A_n(ij)$. Now, for any odd permutation $\tau := \sigma \circ (ij)$, we have

$$\tau(k) = \begin{cases} \sigma(k) & k \neq i, j \\ \sigma(i) & k = j \\ \sigma(j) & k = i \end{cases}.$$

  Finally

$$\begin{aligned} \det(\underline{c}_1, ..., \underline{c}_n) &= \sum_{\sigma \in A_n} a_{\sigma(1)1}... - \sum_{\sigma \in A_n} a_{\tau(1)1}...a_{\tau(i)i}...a_{\tau(j)j}... \\ &= \sum_{\sigma \in A_n} a_{\sigma(1)1}...a_{-} \sum_{\sigma \in A_n} a_{\sigma(1)1}...a_{\sigma(j)i}...a_{\sigma(i)j}... \\ &= \sum_{\sigma \in A_n} a_{\sigma(1)1}...a_{-} \sum_{\sigma \in A_n} a_{\sigma(1)1}...a_{\sigma(j)i}...a_{\sigma(i)i}... \\ &= 0. \end{aligned}$$

□

Corollary. Considering $A \in M_N(\mathbb{F})$ as a list of row vectors $\underline{r}_1, ..., \underline{r}_n \in M_N(\mathbb{F})$, and the function $\det A = \det(\underline{r}_1, ..., \underline{r}_n)$ is

- **multilinear** (linear in each slot separately): if $r_j = \lambda \underline{v} + \mu \underline{w}$ where v, w are column vectors, $\lambda, \mu \in \mathbb{F}$, then $\forall j, 1 \leq j \leq n$

$$\det(\underline{r}_1, ..., \underline{r}_j, ..., \underline{r}_n) = \lambda \det(\underline{r}_1, ..., \underline{v}, ..., \underline{r}_n) + \mu \det(\underline{r}_1, ..., \underline{r}, ..., \underline{r}_n)$$

- **alternating**: if $\underline{r}_i = \underline{r}_j$ for some $i \neq j$, then

$$\det(\underline{r}_1, ..., \underline{r}_n) = 0.$$

Proposition. If $\hat{A}$ is obtained from A by

- $\hat{\underline{c}}_i = \lambda \underline{c}_i$ then $\det \hat{A} = \lambda \det A$
- $\hat{\underline{c}}_i = \underline{c}_i + \lambda \underline{c}_j$ then $\det \hat{A} = \det A$

- $\underline{c}_i \leftrightarrow \underline{c}_j, (i \neq j)$ (swap columns i and j) then $\det \hat{A} = -\det A$

Proof.

- Since $\det(c_1, ..., c_n)$ is multilinear, this comes automatically
- Using multilinearity and alternating properties

$$\begin{aligned}\det(c_1, ..., c_i + \lambda c_j, ..., c_n) &= \det(c_1, ..., c_n) + \lambda \det(c_1, ..., c_j, c_j, ..., c_n) \\ &= \det A + 0.\end{aligned}$$

- If we calculate the determinant of the matrix with cols i and j equal to $c_i + c_j$, then by alternating property it equals zero, and by the multilinear linear property we can show this equals $\det \hat{A} + \det A$.

□

Remark. Since $\det A^T = \det A$, this works just as well for EROs as it does for ECOs. This gives us a method to compute the determinant:

Use guassian elmination (also using ECOs) to turn A into an upper or lower triangular $\hat{A}$, keeping track of the changes to the determinant.

Corollary. If $\hat{A}$ is obtained from A by a mix of EROs and ECOs. Then

$$\det \hat{A} = 0 \Leftrightarrow \det A = 0.$$

Corollary. If $\hat{A}$ is obtained from A by permuting the columns of A (or the rows) by $\sigma \in S_n$ then

$$\det \hat{A} = \text{sign } \sigma \det A.$$

Proof. Write σ as a product of transpositions. Since each swap flips the sign then $\det \hat{A} = \text{sign}(\sigma) \det A$. □

4.3 Characterisation of det

Fix $n > 0$, and let $I := I_n \in M_n(\mathbb{F})$, the identity matrix.

Theorem. Let $D : M_n(\mathbb{F}) \to \mathbb{F}$ be a function that is multilinear and alternating in columns. Then

$$D(A) = D(I) \det A.$$

Proof. Define $c_j = col_j(A) = \sum_{i=1}^n a_{ij} e_i$. Then

$$D(A) = D(c_1, ..., c_n) = \sum_{i_1=1}^n ... \sum_{i_n=1}^n a_{i_1 1} ... a_{i_n n} D(e_{i_1}, ..., e_{i_n})$$

Then because D is alternating, to be nonzero $i_j \neq i_k$ for all j, k, so $\{i_1, ..., i_n\}$ is a permutation of $\{1, ..., n\}$. Therefore we can write it as

$$D(A) = \sum_{\sigma \in S_n} a_{\sigma(1)1} ... a_{\sigma(n)n} D(e_{\sigma(1)}, ..., e_{\sigma(n)}).$$

Since the matrix with columns $e_{\sigma(1)}, ..., e_{\sigma(n)})$ equals I permuted by σ, then $D(...) = \operatorname{sign}(\sigma) D(I)$, thus the final result is obtained. □

Corollary. If $D : M_N(\mathbb{F}) \to \mathbb{F}$ is multilinear and alternating and $D(I) = 1$, then

$$D = \det .$$

Theorem. (Product formula)
If $A, B \in M_n(\mathbb{F})$, then

$$\det(AB) = \det A \det B.$$

Proof. Consider $D : M_n(\mathbb{F}) \to \mathbb{F}, D : B \mapsto \det(AB)$. If D is alternating and multilinear then $D = D(I) \det B = \det A \det B$. Because B could be split into $\lambda B' + \mu B''$ which are identical except for a single column,

$$\det(\lambda AB' + \mu AB'') = \lambda \det AB' + \mu \det AB''.$$

so D is multilinear, and if two columns of B were equal, the columns of AB would also be equal thus $D(B) = 0$ so it is alternating. □

Corollary.

- If $P \in M_m(\mathbb{F})$ is invertible, then $\det(P^{-1}) = (\det P)^{-1}$.
- If A and B are similar, then $\det A = \det B$.

Proof.

- Because $I = PP^{-1}$,

$$1 = \det I = \det P \det P^{-1}$$

- Let $B = P^{-1}AP$, then

$$\det B = \det P^{-1}AP = \det P^{-1} \det A \det P = \det A.$$

□

Remark. If $\phi : V \to V$ is a linear operator and A, B are the matrices of ϕ with respect to two bases, then A and B are similar, so $\det A = \det B$.
Hence the following is well defined.

Definition. Let $\phi : V \to V$ be a linear map, and V be finite dimensional. The **determinant** of ϕ, $\det \phi$, is

$$\det \phi := \det A$$

where A is the matrix of ϕ with respect to some basis.

4.4 Minors, Adjugates and Inverses

We introduce a method for computing the determinant, and the inverse of a matrix.

Definition. For $A = (a_{ij}) \in M_n(\mathbb{F})$ the (i, j)-minor is

$$\mu_{ij}(A) := \det M_{ij}(A)$$

where $M_{ij}(A)$ is the matrix obtained by deleting the ith row and j^{th} column of A.

Proposition. (Column and Row Expansion of Determinant)

- For any fixed column j

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \mu_{ij}(A).$$

- For any fixed row i

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \mu_{ij}(A).$$

Proof. Consider only cols, as by rows can then be proven by taking the transpose. Let $A_j[e_i]$ be the matrix identical to A except $col_j(A) = e_i$. Then

$$A = \sum_{i=1}^{n} a_{ij} A_j[e_i]$$
$$\det A = \sum_{i=1}^{n} a_{ij} \det A_j[e_i].$$

By performing $j - 1$ swaps we can move col j to the first column, and $i - 1$ swaps can move the 1 in e_i to the top, thus making $\hat{A}_j[e_i]$ block upper triangular, so $\det \hat{A}_j[e_i] = \mu_{ij}$ and

$$\det A_j[e_i] = (-1)^{i+j-2} \det \hat{A}_j[e_i] = (-1)^{i+j} \det \hat{A}_j[e_i]$$

$\square$

Definition. The **adjugate** of a matrix $A \in M_N(\mathbb{F})$ is

$$adj(A) = (b_{ij}) \in M_N(\mathbb{F})$$

where

$$b_{ij} = (-1)^{i+j}\mu_{ji}(A)$$

We say $(-1)^{i+j}\mu_{ij}$ is the (i,j)-cofactor and so $adj(A)$ is the transpose of the cofactor matrix.

Theorem.

$$adj(A)A = \det(A)I.$$

Proof. We must show that

$$(ab)_{jk} = \sum_{i=1}^{n} b_{ki}a_{ij} = \begin{cases} \det A & k = j \\ 0 & \text{otherwise} \end{cases}.$$

Since $b_{ki} = (-1)^{k+i}\mu_{ik}(A)$, if $k = j$ the sum equals the determinant using the column expansion formula. If $k \neq j$, then consider $A_k[c_j] = \hat{a}_{ij}$, which is A with column k replaced with column j. Then $\det A_k[c_j] = 0$ as it has two equal columns. By expanding it out on column j

$$\begin{aligned} 0 &= \sum_{i=1}^{n}(-1)^{i+k}\hat{a}_{ik}\mu_{ik}(A_k[c_j]) \\ &= \sum_{i=1}^{n}(-1)^{i+k}a_{ij}\mu_{ik}(A) \\ &= \sum_{i=1}^{n} b_{ik}a_{ij}. \end{aligned}$$

□

Corollary. If $\det A \neq 0$ and then A is invertible and

$$A^{-1} = \frac{1}{\det A}ajd(A).$$

Remark. By the product formula, if A is invertible then $\det A \neq 0$. We also have the converse:

$$A \text{ is invertible } \Leftrightarrow \det A \neq 0.$$

5 Eigenvalues and Diagonalisation

5.1 Fundamental Theorem of Algebra

Theorem. (Fundamental Theorem of Algebra)
Let $f(x) \in \mathbb{C}[x]$ be a non-constant polynomial of degree $d > 0$. The following equivalent statements hold:

- $f(x)$ has at least one complex root
- $f(x)$ has precisely d roots counted with multiplicity
- $f(x)$ factorizes as a product of linear terms

5.2 Eigenvalues and Characteristic Polynomials of Square Matrices

Definition. Let $A \in M_n(\mathbb{F})$. A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there exists a non-zero vector v such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

In this case the vector v is an **eigenvector** of A with **eigenvalue** λ or a **λ-eigenvector**.

Remark. If v is an eigenvector with eigenvector λ then every nonzero scalar multiple av of v is also a λ-eigenvector.

There is a method for computing eigenvectors!

Definition. The characteristic polynomial of a matrix $A \in M_n(\mathbb{F})$ is

$$\Delta_A(t) = \det(A - tI) \in \mathbb{F}[t].$$

Where $I \in M_n(\mathbb{F})$ is the identity matrix $I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & 0 \\ 0 & \dots & 1 \end{pmatrix}$

Proposition. The eigenvalues of A are the roots of $\Delta_A(t)$ in $\mathbb{F}$.

Proof. Assuming A has eigenvalues, we want to solve

$$\begin{aligned} A\mathbf{v} &= \lambda\mathbf{v} \\ (A - \lambda I)\mathbf{v} &= 0 \end{aligned}$$

This only happens when $\ker \phi_{A-\lambda I}$ is nonzero, thus $\phi_{A-\lambda I}$ is not bijective so its determinant equals zero. Thus any eigenvalue is a solution to $\det(A - tI)$ and the eigenvalues are the roots of $\Delta_A(t)$. $\square$

Lemma. If $A \in M_n(\mathbb{F})$ then $\Delta_A(t)$ is a degree n polynomial, with leading

coefficient $(-1)^n$. Therefore $\Delta_A(t)$ has t most n roots. If $\mathbb{F} = \mathbb{C}$ then $\Delta_A(t)$ has precisely n roots counted with multiplicity.

Proof. The leading term of $\Delta_A(t)$ comes from the identity permutation, given by $(a_{11} - t)...(a_{nn} - t)$, thus the leading term will be $(-1)^n t^n$. As it is a degree n polynomial, it has at most n factors in $\mathbb{R}$ and exactly n in $\mathbb{C}$. □

Lemma. Similar matrices have the same characteristic polynomial, hence the same eigenvalues.

Proof. Suppose $A = PBP^{-1}$. Then

$$\begin{aligned}\Delta_A(t) &= \det(A - tI) \\ &= \det(PBP^{-1} - tPP^{-1}) \\ &= \det(P(B - tI)P^{-1}) \\ &= \det P \det(B - tI) \det P^{-1} \\ &= \det(B - tI) = \Delta_B(t).\end{aligned}$$

□

Definition. Let V be a vector space over $\mathbb{F}$, and $\phi : V \to V$ be a linear map.

- $\lambda \in \mathbb{F}$ is an **eigenvalue** of ϕ if there exists nonzero $v \in V$ such that

$$\phi(v) = \lambda v.$$

  Then v is a $\lambda-$**eigenvector** of ϕ.

- The λ-eigenspace of ϕ is

$$\begin{aligned}E_\phi(\lambda) &= \{v \in V \mid \phi(v) = \lambda v\} \\ &= \ker(\phi - \lambda \mathrm{id}_V).\end{aligned}$$

Lemma. Let $A \in M_n(\mathbb{F})$, V be a vector space over $\mathbb{F}$, $\phi : V \to V$ be a linear map. If A is the matrix of ϕ in some basis α then A, ϕ have the same eigenvalues and eigenvectors.

Proof. $\phi = \phi_\alpha \circ \phi_A \circ \phi_\alpha^{-1}$. Thus if v is a $\lambda-$eigenvector of A, then we claim $\phi(v)$ is a $\lambda-$eigenvector of ϕ

$$\begin{aligned}\phi(\phi_\alpha(v)) &= \phi_\alpha \circ \phi_A \circ \phi_\alpha^{-1}(\phi(v)) \\ &= \phi_\alpha(\phi_A(v)) \\ &= \lambda \phi_\alpha(v).\end{aligned}$$

We can prove the converse easily using the fact that $\phi_A = \phi_\alpha^{-1} \circ \phi \circ \phi_\alpha$. □

Definition. The **characteristic polynomial** of $\phi : V \to V$ is

$$\Delta_\phi(t) = \det(\phi - t\mathrm{id}_V).$$

Equivalently, if A represents ϕ in some basis $\Delta_\phi(t) := \Delta_A(t)$.

Remark. The characteristic polynomial ϕ is well defined because similar matrices have equal determinants.

5.3 Diagonalisation

Let $\phi : V \to V$ be a linear map and $A \in M_n(\mathbb{F})$, the matrix of ϕ with respect to some basis $v_1, ..., v_n$. Then the following are equivalent.

- v_i is a λ_i-eigenvector for $i = 1, ..., n$.
- A is diagonal and $A = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$

Definition.

- A linear operator $\phi : V \to V$ is **diagonalisable** if there is a basis consisting of eigenvectors of ϕ. In this case we call it an **eigenbasis**.
- $A \in M_n(\mathbb{F})$ is **diagonalisable** (over $\mathbb{F}$) if there is an invertible $P \in M_n(\mathbb{F})$ such that
$$P^{-1}AP \quad \text{is diagonal.}$$
We say that A is **diagonalised** by P.

Proposition. If $\phi : V \to V$ has a matrix A with respect to a basis β. Then the following are equivalent:

- ϕ is diagonalisable
- ϕ_A is diagonalisable
- A is diagonalisable

Proof. ϕ_A and ϕ have the same eigenvalues and eigenvectors, so one being diagonalisable implies the other is too.
If A is diagonalisable, then

$$\begin{aligned} P^{-1}AP &= D \\ AP &= PD \end{aligned}$$

and since D is diagonal with diagonals $\lambda_1, ..., \lambda_n$, $col_j(PD) = \lambda_j col_j(P)$.

Thus

$$col_j(AP) = \lambda_j col_j(P).$$

Then $A(col_j(P)) = \lambda col_j(P)$, so each column of P is an eigenvector. Since P is invertible its columns are a basis thus ϕ_A is diagonalisable. □

Remark. Many matrices are not diagonalisable over $\mathbb{R}$, but are diagonalisable over $\mathbb{C}$. This happens when the characteristic polynomial has no real roots.

5.4 Applications of Diagonalisation

Diagonalisation can be extremely helpful for efficient calculation of powers of matrices.

Example.

Suppose A is diagonalisable, with $P^{-1}AP = D$ being diagonal. Furthermore P has columns j equal to the λ_j eigenvectors of A. Then $A = PDP^{-1}$, so for $k > 0$

$$\begin{aligned} A^k &= (PDP^{-1})(PDP^{-1})...(PDP^{-1}) \\ &= PD^kP^{-1} \end{aligned}$$

$$\text{with } D^k = \begin{pmatrix} \lambda_1^k & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n^k \end{pmatrix}.$$

This allows us to calculate A^k explicitly with only 2 matrix mutliplications.

5.5 Multiplicity of Eigenvalues and Diagonalisation

We want a necessary and sufficient condition for a matrix or operator to be diagonalisable.

If $\dim V = n$ and $\phi : V \to V$ is linear, then an eigenbasis needs n linearly independent eigenvectors.

Theorem. If $v_1, ..., v_n$ are eigenvectors ϕ with distinct eigenvalues $\lambda_1, ..., \lambda_n$ then $v_1, ..., v_n$ are independent.

Proof. By contradiction, suppose that $a_1v_1 + ... + a_kv_k = 0$ is the smallest dependent list of eigenvectors. Then we have

$$\lambda_1 \sum_{i=1}^{k} a_iv_i = \lambda_1 \cdot 0 = 0$$

$$\phi\left(\sum_{i=1}^{k} a_iv_i\right) = \phi(0) = 0.$$

However $\phi(0)$ also equals

$$\phi\left(\sum_{i=1}^{k} a_i v_i\right) = \sum_{i=1}^{k} a_i \phi(v_i) = \sum_{i=1}^{n} a_i \lambda_i v_i.$$

Then subtracting one from the other means the first term equals zero, meaning there exists a smaller dependence, a contradiction. □

Remark. This can be generalized: if we conbine independent list from each eigenspace we get an independent list.

Corollary. Let $n = \dim V$ and $\phi : V \to V$ be a linear operator. If

$$\Delta_\phi(t) = (t - \lambda_1)...(t - \lambda_n)$$

with λ_i distinct for all i, then ϕ is diagonalisable.

Proof. Because there are n distinct eigenvalues, there are n distinct eigenvectors and since they are independent, they form an eigenbasis. □

Definition. Let λ be an eigenvector of $\phi : V \to V$. Then

- the **algebraic multiplicity** of λ denoted $am(\lambda)$ is the largest k such that $(\lambda - t)^k$ divides $\Delta_\phi(t)$.
- the **geometric multiplicity** of λ is $gm(\lambda) := \dim E_\phi(\lambda)$.

Remark.

- $\dim E_\phi(\lambda)$ is the maximal number of independent $\lambda-$eigenvectors.
- Suppose λ is a root of $\Delta_\phi(t)$. Then $(t - \lambda)$ divides $\Delta_\phi(t)$ so

  $$am(\lambda) \geq 1.$$

  If $E_\phi(\lambda) \neq 0$ then

  $$gm(\lambda) \geq 1.$$

Proposition. For any $\lambda \in \mathbb{F}$ and any linear map ϕ:

$$am(\lambda) \geq gm(\lambda).$$

Proof. Let $v_1, ..., v_k$ be a basis of $E_\phi(\lambda)$. Extend this list to be a basis of V. Then the matrix of ϕ from the standard basis to this basis is diagonal in the top corner, and so if A is the matrix in the bottom corner, then

$$\Delta_\phi(t) = (\lambda - t)^t \det(A - It).$$

Hence $(\lambda - t)$ divides $\Delta_\phi(t)$ at least k times, so $am(\lambda) \geq gm(\lambda)$. $\square$

Theorem. Let $\phi : V \to V$ be a linear operator. Then ϕ is diagonalisable if and only if $\Delta_\phi(t)$ is a product of linear factors and $am(\lambda) = gm(\lambda)$ for all λ.

Proof.
$(\Rightarrow)$ ϕ has an eigenbasis and is diagonalisable, so $\Delta_\phi(t)$ is clearly a product of linear factors. If $am(\lambda) > gm(\lambda)$ then there is a $\lambda-$eigenvector in this eigenbasis that can be written in terms of other $\lambda-$eigenvectors, thus contradicting that ϕ is diagonalisable, thus they are equal.

$(\Leftarrow)$ As $\Delta_\phi(t)$ is a product of linear factors, we can write it as

$$\Delta_\phi(t) = (\lambda_1 - t)^{k_1} ... (\lambda_m - t)^{k_m}$$

and $k_1, ..., k_m$ sum to n. But $k_i = am(\lambda_i) = gm(\lambda_i)$. Thus $\sum_{i=1}^m gm(\lambda_i) = n$. Therefore we have a length n list of eigenvectors and so if they are independent they form a basis and ϕ is diagonalisable.

Suppose there is a dependence. Then we can group all λ_i-eigenvectors together as a single eigenvector w_i. Since there is now a sum of distinct eigenvectors there cannot be a dependence, so the dependence must be inside a w_i. However w_i is formed from a basis so there also cannot be a dependence here. Thus ϕ is diagonalisable.

$\square$

6 Inner Product Spaces

6.1 Inner Products

Recall that the dot (or scalar) product on $\mathbb{R}^n$ is the map

$$\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$$
$$x \cdot y = \sum_{i=1}^n x_i y_i..$$

Then we can define the length of x to be $|x| = x \cdot x$, and the angle between two vectors with $x \cdot y = |x||y| \cos\theta$.

We also have the dot product on $\mathbb{C}^n$, defined as

$$x \cdot y = \sum_{i=1}^n \bar{x}_i y_i.$$

Note that $x\dot{x} = \sum_{i=1}^n \bar{x}_i x_i = \sum_{i=1}^n |x_i|^2$.

Definition. Let V be a vector space over $\mathbb{F}$ which we assume to be either $\mathbb{R}$ or $\mathbb{C}$. An **inner product** on V is a map $V \times V \to \mathbb{F} : (v, w) \mapsto \langle v, w \rangle$ which is

- **congugate symmetric:** $\langle w, v\rangle = \overline{\langle v, w\rangle}$
- **linear in the second argument:**

$$\langle v, \lambda u + \mu w\rangle = \lambda\langle v, u\rangle + \mu\langle v, w\rangle$$

- **positive definite:** For all $v \in V, \langle v, v\rangle \geq 0$ with equality if and only if $v = 0$

A vector space with a fixed inner product is called an **inner product space**.

An inner product space comes with an inner product already defined, denoted $\langle \cdot, \cdot \rangle$.

Remark. $\langle v, v\rangle = \overline{\langle v, v\rangle}$ so the inner product of a vector on itself is always real.

- Suppose $\mathbb{F} = \mathbb{R}$. Then congugate symmetry is just symmetry, and so $\langle v, w\rangle = \langle w, v\rangle$. Furthermore this means the inner product is lienar in both arguments.
- Now suppose $\mathbb{F} = \mathbb{C}$. Then

$$\langle \lambda u + \mu v, w\rangle = \bar{\lambda}\langle u, w\rangle + \bar{\mu}\langle v, w\rangle.$$

Definition. Let v be an inner product space over $\mathbb{R}$.

- The **norm** of a vector v is $\|v\| = \sqrt{\langle v, v\rangle} \geq 0$.

  If $\|v\| = 1$ then we say v is **unit-length**, and $\frac{v}{\|v\|}$ is the **normalisation** of v.
- Let $v, w \in V$. If $\langle v, w\rangle = 0$, then v is **orthogonal** to w, written $v \perp w$.

6.2 Cauchy-Schwarz Inequality

Theorem. Let V be an inner product space. For $v, w \in V$,

$$|\langle v, w\rangle| \leq \|v\| \, \|w\|$$

with equality if and only if v and w are dependent.

Proof. If $v = 0$ both terms equal 0, otherwise let $w = \lambda v + u$, with $v \perp u$. Then

$$\begin{aligned}\langle v, \lambda v + u\rangle &= \lambda\langle v, v\rangle + \langle v, u\rangle \\ &= \lambda \|v\|^2\end{aligned}$$

Thus $\lambda = \frac{\langle v,w\rangle}{\|v\|^2}$. Then using the positive definite property gives

$$\begin{aligned} 0 \leq \|u\|^2 &= \langle w - \lambda v, u\rangle \\ &= \langle w, u\rangle - \lambda\langle v, u\rangle \\ &= \langle w, w - \lambda v\rangle \\ &= \|w\|^2 - \lambda\langle w, v\rangle \\ &= \|w\|^2 - \frac{\langle v, w\rangle\langle w, v\rangle}{\|v\|^2} \\ |\langle v, w\rangle|^2 &\leq \|v\|^2 \|w\|^2 \end{aligned}$$

and we can take the square root for the final result. Then if there is equality, $\|u\|^2 = 0$ so $w = \lambda v$ thus they are dependent. □

Corollary. (Triangle Inequality) For $v, w \in V$ we have

$$\|v + w\| \leq \|v\| + \|w\|$$

with equality if and only if v and w are dependent.

Proof. By expanding out $\langle v + w, v + w\rangle$ it can be shown

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w\rangle^2 \\ &= \|v\|^2 + \|w\|^2 + \langle v, w\rangle + \overline{\langle v, w\rangle} \\ &= \|v\|^2 + \|w\|^2 + 2Re(\langle v, w\rangle). \end{aligned}$$

Then $Re(\langle v, w\rangle) \leq |\langle v, w\rangle| \leq \|v\| \|w\|$ so

$$\begin{aligned} \|v + w\|^2 &\leq \|v\|^2 + 2\|v\| \|w\| \|w\|^2 \\ &= (\|v\| + \|w\|)^2 \end{aligned}$$

and the final result can be obtained by taking the square root. □

6.3 Orthogonality

Definition. A list $u_1, ..., u_k$ in an inner product space V is **orthonormal** if for $1 \leq i, j \leq k$

$$\langle u_i, u_j\rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Remark. A list is orthonormal if every u_i has unit length and distinct u_i, u_j are orthogonal.

Lemma. For an orthonormal basis $u_1, ..., u_n$ in an inner product space V,

for a vector $v \in V$ we have

$$v = \sum_{i=1}^{n} \langle u_i, v \rangle u_i.$$

Proof. Since $u_1, ..., u_n$ are a basis $v = \sum_{i=1}^{n} \lambda_i u_i$. Furthermore

$$\begin{aligned} \langle u_i, v \rangle &= \sum_{j=1}^{n} \lambda_j \langle u_i, u_j \rangle \\ &= 0 + ... + \lambda_i + ... + 0. \end{aligned}$$

Thus if every $\lambda_i = \langle u_i, v \rangle$ then the final result is obtained. □

Lemma. Any orthonormal list is linearly independent.

Proof. If $v = 0$, then $\langle u_i, v \rangle = 0$ so all $\lambda_i = 0$ so the list must be independent. □

Proposition. If $u_1, ..., u_n$ is an orthonormal basis for V and

$$\begin{aligned} v &= \sum_{i=1}^{n} x_i u_i \\ w &= \sum_{i=1}^{n} y_i u_i \end{aligned}$$

then

$$\langle v, w \rangle = \sum_{i=1}^{n} \bar{x}_i y_i = x \cdot y$$

Proof. Expand using sesquilinearity.

$$\begin{aligned} \langle v, w \rangle &= \langle \sum_{i=1}^{n} x_i u_i, \sum_{j=1}^{n} y_j u_j \rangle \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} \bar{x}_i y_i \langle u_i, u_j \rangle \\ &= \sum_{i=1}^{n} \bar{x}_i y_i. \end{aligned}$$

□

Theorem. (Gram-Schmidt Orthogonalisation) Let $v_1, ..., v_n$ be an independent list in an inner product space V. Then there exists an orthonormal

list $u_1, ..., u_n$ such that for all $1 \leq k \leq u$ we have

$$span\ \{v_1, ..., v_k\} = span\ \{u_1, ..., u_k\}.$$

We construct $u_1, ..., u_k$ inductively:

- For $k = 1$ we set $w_1 = v_1$, and $u_1 = \frac{w_1}{\|w_1\|}$
- For $k > 1$, suppose we have defined $u_1, ..., u_k$ already and set

$$\begin{aligned} w_k &= v_k - \sum_{j=1}^{k-1} \langle u_j, v_k \rangle u_j \\ &= v_k - \sum_{j=1}^{k-1} \frac{\langle w_j, v_k \rangle}{\|w_j\|^2} w_j. \end{aligned}$$

  and $u_k = \frac{w_k}{\|w_k\|}$.

Proof. Using induction on k, we see the base case is trivial. Then given $u_1, ..., u_{k-1}$ orthogonal with span equal to that of $v_1, ..., v_{n-1}$. Then

$$\begin{aligned} \text{span}\{u_1, ..., u_k\} &= \text{span}\{v_1, ..., u_k\} = \text{span}\{v_1, ..., w_k\} \\ &= \text{span}\{v_1, ..., v_k - \sum_{i=1}^{n} \langle u_j, v_k \rangle u_j\} \\ &= \text{span}\{v_1, ..., v_n\}. \end{aligned}$$

Next, we need to show w_k is orthogonal to all u_i, $i < k$. Then

$$\begin{aligned} \langle u_i, w_k \rangle &= \langle u_i, v_k - \sum_{i=1}^{k-1} \langle u_i, v_k \rangle u_j \rangle \\ f &= \langle u_i, v_k \rangle - \sum_{j=1}^{k-1} \langle u_j, v_k \rangle \langle u_i, u_j \rangle \\ f &= \langle u_i, v_k \rangle - \langle u_i, v_k \rangle = 0. \end{aligned}$$

□

Corollary. Any finite dimensional inner product space has an orthonormal basis.

Proof. Apply Gram-Schmidt orthogonalisation to a basis of V. □

Remark. In practise it is easier to start with $v_1, ..., v_n$, then product an $w_1, ..., w_n$ satisfying $w_1 \perp w_j (i \neq j)$ then finally set $u_i = \frac{w_i}{\|w_i\|}$ for every i to get an orthonormal list.

Definition. A matrix $Q \in M_n(\mathbb{R})$ is orthogonal if $Q^T Q = \mathbb{I}_n$

Remark. Q is orthogonal if and only if the columns of Q are orthonormal with respect to the dot product.

Theorem. (QR Decomposition)
Let $A \in M_n(\mathbb{R})$ be invertible. Then there exists orthogonal Q and upper-triangular R such that

$$A = QR.$$

Proof. Let A have columns $v_1, ..., v_n$. Then as A is invertible these are a basis of $\mathbb{R}^n$. Applying the Gram-Schmidt process gives and orthonormal basis $u_1, ..., u_n$. Then

$$\begin{aligned} v_1 &= \|w_1\| \, u_1 \\ v_2 &= \|w_2\| \, u_2 + \langle u_1, v_2 \rangle u_1 \\ &\dots \\ v_k &= \|w_k\| \, v_k + \sum_{j=1}^{k} \langle u_j, u_k \rangle u_j. \end{aligned}$$

Then $u_1, ..., u_n$ define the columns of Q and let

$$R_{ij} = \begin{cases} \langle u_i, v_j \rangle & i < j \\ \|w_i\| & i = j \\ 0 & i > j \end{cases}.$$

Then $A = QR$. □

Remark. To compute Q and R in practise it is simplest to first find Q, then at $Q^T = Q^{-1}$, we have

$$R = Q^T A.$$

6.4 Orthogonal Complements

Let V be an inner product space and let $U \leq V$. Recall $u, v \in V$ are orthogonal, $u \perp v$, if $\langle u, v \rangle = 0$.

Definition. The **orthogonal complement** $U^\perp$ of u in V is

$$U^\perp = \{v \in V \mid \langle u, v \rangle = 0 \text{ for } u \in U\}.$$

Proposition.

- $U^\perp$ is a subspace of V
- $U \cap U^\perp = \{0\}$
- $U \subseteq (U^\perp)^\perp$

Proof.

- From the linearity in the second argument
- Let $u \in U$ and $u \in U^\perp$. Then $\langle u, u\rangle = 0$ which implies $u = 0$ by the positive definite axiom.
- If $u \in U, w \in U^\perp$ then

$$\langle w, u\rangle = \overline{\langle u, w\rangle} = 0$$

so $u \in (U^\perp)^\perp$.

□

Theorem. If $U \leq v$ is a finite dimensional subspace then $V = U + U^\perp$.

Proof. It's sufficient to show for any $v \in V$ there exists $v_1 \in U$ and $v_2 \in U^\perp$ such that $v = v_1 + v_2$.
Let $u_1, ..., u_k$ be an orthonormal basis of U. Given $v \in V$ set

$$\begin{aligned} v_1 &= \sum_{i=1}^{k} \langle u_i, v\rangle u_i \\ v_2 &= v - v_1. \end{aligned}$$

Then $v_1 \in U$ and to show $v_2 \in U^\perp$:

$$\begin{aligned} \langle u_j, v_2\rangle &= \langle u_j, v\rangle - \langle u_j, v_1\rangle \\ &= \langle u_j, v\rangle - \sum_{i=1}^{k} \langle u_i, v\rangle \langle u_j, u_j\rangle \\ &= \langle u_j, v\rangle - \langle u_j, v\rangle = 0. \end{aligned}$$

□

Corollary. Let V be a finite dimensional inner product space.

- $\dim U^\perp = \dim V - \dim U$
- $U = (U^\perp)^\perp$

Proof.

- As $V = U + U^\perp$, we have

$$\begin{aligned}\dim V &= \dim U + \dim U^\perp - \dim(U \cap U^\perp) \\ &= \dim U + \dim U^\perp - 0.\end{aligned}$$

- $U \leq (U^\perp)^\perp$. As they have the same rank they are equal.

□

7 Linear Operators on Inner Product Spaces

Continue to assume that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

7.1 Adjoints

Let V be an inner product space.

Notation. $L(V) := \{f : V \to V \mid f \text{ linear}\}$.

Note that $L(V)$ is itself a vector space.

Lemma. (Nondegeneracy Lemma)
For $v \in V$ the following are equivalent.

- $\langle v, V\rangle = \{0\}$
- $v = 0$.

Proof.

- ($\Leftarrow$). Let $v = 0$, then for $u \in U$

$$\langle v, u\rangle = \overline{\langle u, v\rangle} = 0.$$

- ($\Rightarrow$). Let $\langle v, V\rangle = \{0\}$. Then

$$\langle v, v\rangle = 0$$

so by the positive definite axiom $v = 0$.

□

Definition. Let $\phi \in L(V)$. An **adjoint** to ϕ is as operator $\phi^* \in L(V)$ such that for $v, w \in V$

$$\langle \phi^*(v), u\rangle = \langle u, \phi(w)\rangle.$$

Equivalently,

$$\langle w, \phi^*(v)\rangle = \langle \phi(w), v\rangle.$$

Proposition. Let $\dim V < \infty$ and $\phi \in L(V)$. Then

- ϕ has a unique adjoint ϕ^*
- Let $\alpha : u_1, ..., u_n$ be an orthonormal basis of V with respect to which ϕ has matrix $A = (a_{ij})$. Then ϕ^* has matrix

$$A^\dagger := \bar{A}^T.$$

Proof.

- To prove uniqueness, we first need to prove that every ϕ which acts as we expect, and that if two maps are both adjoint to ϕ they must be equal. In an inner product space with orthonormal basis $u_1, ..., u_n$ any $u \in V$ can be written as

$$u = \sum_{i=1}^{n} \langle u_i, u \rangle u_i.$$

In particular if ϕ^* is adjoint to ϕ then

$$\begin{aligned} \phi^*(v) &= \sum_{i=1}^{n} \langle u_i, \phi^*(v) \rangle u_i. \\ &= \sum_{i=1}^{n} \langle \phi(u_i), v \rangle u_i \end{aligned}$$

so ϕ^* is uniquely determined by ϕ. Now we need to prove it has the necessary properties. ϕ^* is linear because of linearity in the second slot. Furthermore for $w \in V$

$$w = \sum_{i=1}^{n} \langle u_i, w \rangle u_i$$

so

$$\phi(w) = \sum_{i=1}^{n} \langle u_i, w \rangle \phi(u_i).$$

Therefore

$$\begin{aligned} \langle \phi^*(v), w \rangle &= \sum_{i=1}^{n} \overline{\langle \phi(u_i), v \rangle} \langle u_i, w \rangle. \\ &= \sum_{i=1}^{n} \langle v, \phi(u_i) \rangle \langle u_i, w \rangle \\ &= \langle v, \sum_{i=1}^{n} \langle u_i, w \rangle \phi(u_i) \rangle = \langle v, \phi(w) \rangle. \end{aligned}$$

Now suppose ϕ_1^* and ϕ_2^* are both adjoint to ϕ. Then the nondegeneracy lemma can show $\phi_1^* - \phi_2^* = 0$, so they are equal.

- For an arbitary linear map ψ with matrix $B = (b_{ij})$, we have

$$\langle u_k, \psi(u_j)\rangle = \sum_{i=1}^{n} b_{ij}\langle u_k, u_i\rangle = b_{kj}.$$

Then, for ϕ^*, we have

$$\begin{aligned} c_{kj} = \langle u_k, \phi^*(u_j)\rangle &= \langle \phi(u_k), u_j\rangle \\ &= \sum_{i=1}^{n} \overline{a_{ik}}\langle u_i, v_j\rangle \\ &= \overline{a_{jk}}. \end{aligned}$$

□

Corollary. Let $A \in M_n(\mathbb{F})$. Then

$$\begin{aligned} (\phi_A)^* &= \phi_{\bar{A}^T} \quad \text{for } \mathbb{F} = \mathbb{C} \\ (\phi_A)^* &= \phi_{A^T} \quad \text{for } \mathbb{F} = \mathbb{R}. \end{aligned}$$

Proposition. Let $\phi, \psi \in L(V)$. Then

- $(\phi \circ \psi)^* = \psi^* \circ \phi^*$
- $(\phi + \lambda\psi)^* = \phi^* + \bar{\lambda}\psi^*$
- $(\phi^*)^* = \phi$
- $\mathrm{id}_V^* = \mathrm{id}_V$

Definition. Let V be an inner product space.

- Let $\phi \in L(V)$.
  - ϕ is **self-adjoint** if $\phi = \phi^*$
  - ϕ is **skew-adjoint** if $\phi = -\phi^*$
- Let $A \in M_n(\mathbb{F})$. If $\mathbb{F} = \mathbb{C}$
  - We say A in **Hermitian** if

$$A = A^\dagger = \bar{A}^T.$$

  - We say A is **skew-Hermitian** if

$$A = -A^\dagger = -\bar{A}^T.$$

  If $\mathbb{F} = \mathbb{R}$:

– We say A is **symmetric** if

$$A = A^T.$$

– We say A is **skew-symmetric** if

$$A = -A^T.$$

Remark. For $\phi \in L(V)$ and $A \in M_n(\mathbb{F})$

$$\phi \text{ is self-adjoint } \Leftrightarrow A \text{ is Hermitian or symmetric.}$$

Similarly,

$$\phi \text{ is skew-adjoint } \Leftrightarrow A \text{ is skew-Hermitian or skew-symmetric.}$$

Example.

- $\mathrm{id}_V^* = \mathrm{id}_V$ so id_V is self adjoint
- $\phi \circ \phi^*$ is self-adjoint

$$(\phi \circ \phi^*)^* = (\phi^*)^* \circ \phi^* = \phi \circ \phi^*.$$

- for $\mathbb{F} = \mathbb{C}$

$$\phi \text{ is self-adjoint } \Leftrightarrow i\phi \text{ is skew-adjoint.}$$

Definition.

- We say $\phi \in L(V)$ is an **orthogonal transformation** if $\phi^* = \phi^{-1}$ and $\mathbb{F} = \mathbb{R}$
- We say $\phi \in L(V)$ is an **unitary transformation** if $\phi^* = \phi^{-1}$ and $\mathbb{F} = \mathbb{C}$
- We say $A \in M_n(\mathbb{F})$ is **orthogonal** if

$$A^T A = \mathrm{id} \quad \text{and} \quad \mathbb{F} = \mathbb{R}.$$

- We say $A \in M_n(\mathbb{F})$ is **unitary** if

$$\bar{A}^T A = \mathrm{id} \quad \text{and} \quad \mathbb{F} = \mathbb{C}.$$

Remark. Let ϕ have matrix A with respect to an orthormal basis. ϕ is orthogonal or unitary iff A is orthogonal or unitary.

7.2 The Spectral Theorem

Definition. Let $\phi \in L(V)$. Then $U \leq V$ is **ϕ-invariant** if $\phi(U) \leq U$. In other words $\phi(u) \in U$ for $u \in U$.

Lemma. If $\phi, \psi \in L(v)$ and

$$\phi \circ \psi = \psi \circ \phi$$

then $E_\phi(\lambda)$ is $\psi-$invariant for all $\lambda \in \mathbb{F}$.

Proof. Let λ be an eigenvalue of ϕ and $v \in E_\phi(\lambda)$. Then

$$\begin{aligned} \phi(v) &= \lambda v \\ \psi(\phi(v)) &= \psi(\lambda v) \\ \phi(\psi(v)) &= \lambda \psi(v) \end{aligned}$$

so $\psi(v) \in E_\phi(\lambda)$, thus $E_\phi(\lambda)$ is $\psi-$invariant. $\square$

Lemma. Let V be finite dimensional and $\phi \in L(V)$. If $U \leq V$ is $\phi-$invariant then $U^\perp$ is ϕ^*-invariant.

Proof. Let $v \in U^\perp$ so that for any $u \in U$, $\langle u, v \rangle = 0$.

$$\langle u, \phi^*(v) \rangle = \langle \phi(v), v \rangle = 0.$$

$\square$

Definition. Let V be a finite dimensional inner product space. A linear operator ϕ is **normal** if it commutes with its adjoint:

$$\phi \circ \phi^* = \phi^* \circ \phi.$$

Example.

- Self adjoint and skew-adjoint operators are normal
- Unitary and orthogonal transformations are normal.

Proposition. Let V be finite dimensional and $\phi \in L(V)$. If ϕ is normal then $E_\phi(\lambda)^\perp$ is $\phi-$invariant.

Proof. Since ϕ and ϕ^* commute, $E_\phi(\lambda)$ is ϕ^*-invariant. Then $E_\phi(\lambda)^\perp$ is $(\phi^*)^*-$invariant. $\square$

Definition. Let V be finite dimensional. A linear operator $\phi \in L(V)$ is **orthogonally diagonalisable** if V admits an orthonormal basis of eigenvectors for ϕ.

Proposition. Let V be finite dimensional over $\mathbb{F}$. Let $\phi \in L(V)$ be orthogonally diagonalisable. Then ϕ is normal, and if $\mathbb{F} = \mathbb{R}$ then ϕ is self-adjoint.

Proof. ϕ has matrix A with respect to this orthonormal basis, which is diagonal with $a_{ii} = \lambda_i$. Then $(a^\dagger)_{ii} = \bar{\lambda}_i$ and $A^\dagger A = AA^\dagger$ as diagonal matrices commute. Hence ϕ is normal and if $\mathbb{F} = \mathbb{R}$ ϕ is self-adjoint. $\square$

Theorem. (Spectral Theorem over $\mathbb{C}$)
Let V be a finite-dimensional inner product space of dimension n over $\mathbb{C}$, and let $\phi \in L(V)$.

- ϕ is orthogonally diagonalisable if and only if ϕ is normal
- If $A \in M_n(\mathbb{C})$ is Hermitian then there is a unitary matrix P where $P^{-1}AP$ which is diagonal.

Proof.

- For the first part we proceed by induction on n. If $n = 1$ this is trivial. Now suppose for a vector space U with $\dim U < n$, $\phi \in L(U)$ is orthogonally diagonalisable if it is normal.

  As $\mathbb{F} = \mathbb{C}$, we know ϕ has at least one eigenvalue, λ, so $E_\phi(\lambda)$ is nonempty. Therefore $E_\phi(\lambda)^\perp$ has dimension less than V. It is $\phi-$invariant since ϕ is normal and, as $E_\phi(\lambda)$ is $\phi-$invariant, $E_\phi(\lambda)^\perp$ is also ϕ^*-invariant.

  Therefore the restriction of ϕ to $E_\phi(\lambda)^\perp$ is normal and we can find an orthonormal basis of $\phi-$eigenvectors. Combining this with an orthonormal basis of $E_\phi(\lambda)$ we obtain an orthogonal basis of ϕ-eigenvectors.

- $\phi_A : \mathbb{C}^n \to \mathbb{C}^n$ is self-adjoint and thus normal with respect to the dot-product, so it is orthogonally diagonalisable with basis $u_1, ..., u_n$. Let P have columns $u_1, ..., u_n$ which is an eigenbasis, so $P^{-1}AP$ is diagonal, and P is unitary because its columns are orthonormal.

$\square$

Now, to prove the Spectral Theorem over $\mathbb{R}$ we need a way to guarantee a matrix has an eigenvector.

Lemma. Let V be a complex inner product space and let $\phi \in L(V)$ be self-adjoint.

- Every eigenvalue of ϕ is real
- If u, v are eigenvectors of eigenvalues λ, μ with $\lambda \neq \mu$ then $u \perp v$.

Proof. Since ϕ is self-adjoint

$$\langle \phi(v), w\rangle = \langle v, \phi(w)\rangle$$
$$\bar{\lambda}\langle v, w\rangle = \mu\langle v, w\rangle$$

so $(\bar{\lambda} - \mu)\langle v, w\rangle = 0$. Then

$$(\bar{\lambda} - \lambda)\,\|v\|^2 = 0$$

but v is non-zero as it is an eigenvector so $\bar{\lambda} = \lambda$ and $\lambda \in \mathbb{R}$. Then $(\lambda - \mu)\langle v, w\rangle = 0$ and as $\lambda \neq \mu$ they must be orthogonal. □

Proposition. A self-adjoint operator on a real finite dimensional inner product space V has a real eigenvalue.

Proof. Let A be the matrix of ϕ, and so $A^T = A$. Now consider $\phi_A^{\mathbb{C}}$ as the complex linear operator of A. Then as $\phi_A^{\mathbb{C}}$ is self-adjoint, so it has a real eigenvalue λ. Then

$$0 = \Delta_{\phi_A^{\mathbb{C}}}(\lambda) = \det(A - \lambda I) = \Delta_\phi(\lambda)$$

so ϕ has a real eigenvector. □

Remark. The requirements for ϕ to be diagonalisable over $\mathbb{R}$ are stronger because we need stronger requirements to guarantee an eigenvalue exists.

Theorem. (Spectral Theorem over $\mathbb{R}$)
Let V be a finite-dimensional inner product space of dimension n over $\mathbb{F}$, and let $\phi \in L(V)$.

- ϕ is orthogonally diagonalisable if and only if ϕ is self-adjoint.
- If $A \in M_n(\mathbb{R})$ is symmetric then there is an orthogonal matrix P where $P^{-1}AP$ is diagonal.

Proof. The proof is identical except that we obtain an eigenvalue using the fact that ϕ is self-adjoint instead. □